

Lie Theory for the Impatient

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Abstract

This document collects a variety of definitions and results from Lie theory. To avoid distraction, all vector spaces are over $\mathbb{C}$, and all manifolds are real.

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1 Lie Algebras

1.1 Starting Out

Definition 1 (Lie algebra). A Lie algebra is a vector space $\mathfrak{g}$ equipped with a Lie bracket $[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying the following properties.

- (i) Bilinear: the map $[-, -]$ is bilinear.
- (ii) Alternating: $[X, X] = 0$ for all $X \in \mathfrak{g}$.
- (iii) Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ for all $X, Y, Z \in \mathfrak{g}$.

A morphism of Lie algebras is a linear map preserving the bracket.

Example 2. For any (possibly non-commutative, non-unital) algebra A over $\mathbb{C}$, one can define the Lie bracket $[-, -]: A \times A \rightarrow A$ by

$$[a, b] := ab - ba.$$

For example, applying this to $A = \text{End}(V)$ where V is a vector space gives the Lie algebra $\mathfrak{gl}(V)$. If further $V = \mathbb{C}^n$, we abbreviate $\mathfrak{gl}(V)$ to $\mathfrak{gl}_n(\mathbb{C})$.

Example 3. The subspace

$$\mathfrak{sl}_n(\mathbb{C}) := \{X \in \mathfrak{gl}_n(\mathbb{C}) : \text{tr } X = 0\}$$

is a Lie subalgebra.

Example 4. The subspace of alternating matrices is a Lie subalgebra, called $\mathfrak{o}_n(\mathbb{C}) \subseteq \mathfrak{gl}_n(\mathbb{C})$.

Definition 5 (representation). A representation of a Lie algebra $\mathfrak{g}$ is a vector space V along with a morphism $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$. A morphism of representations is a morphism of the underlying vector spaces which preserves the $\mathfrak{g}$ -action.

Example 6 (adjoint). The adjoint representation of a Lie algebra $\mathfrak{g}$ is the morphism $\text{ad}_\bullet: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ defined by

$$\text{ad}_X(Y) := [X, Y].$$

The fact that $\text{ad}_\bullet$ preserves the Lie bracket is equivalent to the Jacobi identity.

Definition 7 (irreducible). A representation $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is *irreducible* if and only if $V \neq 0$, and V admits no nonzero proper subrepresentations. A representation is *completely reducible* if and only if it is a direct sum of irreducible representations.

Example 8. Let $\mathfrak{b} \subseteq \mathfrak{gl}_2(\mathbb{C})$ be the Lie subalgebra of upper-triangular matrices. Then the representation $\mathfrak{b} \rightarrow \mathfrak{gl}_2(\mathbb{C})$ is not completely reducible.

Lemma 9 (Schur). Let $\rho_V: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ and $\rho_W: \mathfrak{g} \rightarrow \mathfrak{gl}(W)$ be representations, and let $\varphi: V \rightarrow W$ be a nonzero morphism of representations.

- (a) If V is irreducible, then φ is injective.
- (b) If W is irreducible, then φ is surjective.
- (c) If both V and W are irreducible, then φ is an isomorphism.

Sketch. Here, (c) follows from (a) and (b). We only sketch (a): because φ is nonzero, $\ker \varphi$ is a proper subrepresentation of V , so $\ker \varphi = 0$ because V is irreducible. ■

Definition 10 (universal enveloping algebra).